KR-ACC-24 Access Node QoS Congestion Event Resiliency SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5620

Description

The KR-ACC-24 node failure, situated at KR Puram Access, is an instance of a QoS congestion event precipitated by microbursts and interface tail drops. This phenomenon is a direct consequence of the Arista Access node's QoS queue starvation, stemming from an imbalance in traffic prioritization and policing across the various egress queues. The underlying protocol mechanism, responsible for this issue, is the Arista's egress queuing architecture, which allocates bandwidth to queues based on configured policies. When the egress queues become congested due to excessive traffic, the egress queuing architecture fails to allocate sufficient bandwidth to the affected queues, resulting in tail drops and microbursts. As the congestion persists, it leads to QoS queue starvation, further exacerbating the issue. To mitigate this phenomenon, it is imperative to identify and rectify the root cause of the congestion, which may be attributed to various factors, including misconfigured QoS policies, inadequate policing, or excessive traffic loads.

Pre-requisite Checks

- Verify that the KR Puram Access site is accessible and that all necessary personnel and equipment are present.

- Confirm that the redundant link is functioning correctly and that there are no issues with the backup power supply.

- Perform a preliminary investigation to determine if the issue is related to a specific interface or a network-wide problem.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate policing	Medium
Excessive traffic loads	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 is up, line protocol is up (connected), Description:
This command is used to verify the operational status of the interface
and confirm whether it is up and functioning correctly.

Command: show qos policy KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 QoS Policy: policy-name (policy-id), Description: This
```

```
command is used to display the QoS policy configuration for the
interface and confirm whether it is correctly applied.

Command: show policer KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 Policer: policer-name (policer-id), Description: This
command is used to display the policer configuration for the interface
and confirm whether it is correctly applied.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy KR-ACC-24-1/1/1, Description: Verify the QoS policy configuration for the interface and confirm whether it is correctly applied. If the policy is misconfigured, rectify the issue by modifying the policy configuration.

Step 2: Step: Verify policer configuration, Command: show policer KR-ACC-24-1/1/1, Description: Verify the policer configuration for the interface and confirm whether it is correctly applied. If the policer is misconfigured, rectify the issue by modifying the policer configuration.

Step 3: Step: Verify traffic loads, Command: show interface KR-ACC-24-1/1/1, Description: Verify the traffic loads on the interface to determine whether they are within normal operating limits. If the traffic loads are excessive, consider implementing traffic shaping or other traffic management techniques to mitigate the issue.

Post-Check Verification

```
Command: show interface KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 is up, line protocol is up (connected), Description:
Verify the operational status of the interface and confirm whether it
is up and functioning correctly.

Command: show qos policy KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 QoS Policy: policy-name (policy-id), Description:
Verify the QoS policy configuration for the interface and confirm
whether it is correctly applied.

Command: show policer KR-ACC-24-1/1/1, Expected Output:
KR-ACC-24-1/1/1 Policer: policer-name (policer-id), Description:
Verify the policer configuration for the interface and confirm whether
it is correctly applied.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure replace KR-ACC-24-1/1/1 qos policy original, Description: Restore the original QoS policy configuration for the interface to ensure that the issue is mitigated.

- Step: Restore original policer configuration, Command: configure replace KR-ACC-24-1/1/1 policer original, Description: Restore the original policer configuration for the interface to ensure that the issue is mitigated.